Skincare Product Demand Forecast

Full Year 2026 — Analytics & Strategy Summary Report

38,399 Total 2026 Forecast Units	+1.10% YoY Growth vs 2025	$344,192 3yr Lost Revenue	20 SKUs 8 Categories	11,762 Units Lost to Stockout

1. Executive Summary

This report presents the complete 2026 annual demand forecast for a 20-SKU skincare portfolio spanning 8 product categories. The forecast corrects for a critical bias in traditional forecasting: **stockout suppression** — where Sales Qty is capped by available stock during out-of-stock periods, causing systematic under-forecasting of high-demand SKUs. All models use true Demand Qty as input.

The 2026 blended forecast totals **38,399 units** — a **+1.10% increase** over 2025 demand (37,982 units). The model blends 60% Weighted Moving Average (WMA) with 40% Holt's Double Exponential Smoothing (DES) to capture both recent trajectory and demand momentum. Makeup and Cleanser categories lead growth, while Treatment and Care face mild contraction.

2. Forecasting Methodology

Step	Method	Description
1	Data Foundation	720 rows: 20 SKUs x 3 years x 12 months. Uses Demand Qty, not Sales Qty.
2	Stockout Correction	Lost Units = Demand - Sales. True demand recovered to remove suppression bias.
3	Seasonality Index (SI)	SI = Mean Monthly Demand / Mean All-Month Demand per SKU. 3yr average.
4	WMA Growth	60% weight on 2024→25 growth + 40% on 2023→24. Cap: -20% / +25%.
5	Holt's DES	Level (alpha=0.50) + Trend (beta=0.30) smoothing. Projects level + trend momentur
6	Blend	Final = WMA x 0.60 + Holt's x 0.40. Blending reduces model-specific error.
7	Monthly Split	Monthly Fcst = Annual Blend x SI / 12. Preserves seasonal pattern.
8	Safety Stock	SS = 1.645 x sigma x sqrt(Lead Time). 95% service level. Min floor = 10 units.
9	Reorder Point	ROP = Avg Monthly Demand x Lead Time + Safety Stock. Lead time = 1 month.

3. Category-Level Forecast Summary

Category	SKUs	2023	2024	2025	2026F	Growth%	Avg Stockout	Avg Fill Rate
Treatment	6	11,657	10,839	11,416	10,980	-3.8%	72.7%	89.1%
Hydration	3	5,596	5,489	5,798	5,795	-0.1%	78.7%	87.6%
Care	3	5,515	5,286	5,495	5,379	-2.1%	68.5%	89.3%
Makeup	2	3,592	3,532	4,215	4,369	+3.7%	72.2%	89.9%
Mask	2	3,414	3,728	3,705	3,957	+6.8%	68.1%	91.1%
Cleanser	2	3,322	3,660	3,569	3,827	+7.2%	63.9%	89.8%

Exfoliation	1	1,646	1,699	1,963	2,079	+5.9%	77.8%	87.2%
Protection	1	1,658	1,923	1,821	2,013	+10.5%	63.9%	92.0%
TOTAL	**20**	**36,400**	**36,156**	**37,982**	**38,399**	**+1.1%**	**—**	**—**

4. Full SKU Demand Forecast Detail

SKU	Product	Cat	2023	2024	2025	2026F	Growth%	Trend	SafeSt	ROP
SK001	Face Wash	Clnsr	1,691	1,864	1,681	1,782	+6.0%	Stable	64	209
SK002	Moisturizer	Hydr	1,951	1,811	1,844	1,744	-5.4%	Stable	69	225
SK003	Sunscreen SPF50	Prot	1,658	1,923	1,821	2,013	+10.5%	Stable	75	225
SK004	Night Cream	Trtmt	1,897	1,668	2,101	2,052	-2.3%	Growing	71	228
SK005	Serum Vit C	Trtmt	1,949	1,903	1,837	1,788	-2.7%	Stable	75	233
SK006	Anti-Aging Cream	Trtmt	2,041	1,895	1,867	1,750	-6.3%	Stable	81	242
SK007	Face Toner	Clnsr	1,631	1,796	1,888	2,045	+8.3%	Growing	59	207
SK008	Body Lotion	Hydr	1,847	1,886	1,945	1,990	+2.3%	Stable	72	230
SK009	Lip Balm	Care	1,938	1,805	1,695	1,573	-7.2%	Declining	69	220
SK010	Eye Cream	Trtmt	1,933	1,701	1,709	1,536	-10.1%	Declining	61	209
SK011	Acne Gel	Trtmt	1,838	1,672	2,049	2,028	-1.0%	Growing	85	239
SK012	Sheet Mask	Mask	1,767	1,861	1,769	1,824	+3.1%	Stable	74	224
SK013	Face Scrub	Exfol	1,646	1,699	1,963	2,079	+5.9%	Growing	77	224
SK014	BB Cream	Mkup	1,762	1,628	2,049	2,071	+1.1%	STRONG	75	226
SK015	CC Cream	Mkup	1,830	1,904	2,166	2,298	+6.1%	Growing	76	240
SK016	Hand Cream	Care	1,722	1,631	1,837	1,819	-1.0%	Growing	86	230
SK017	Foot Cream	Care	1,855	1,850	1,963	1,987	+1.2%	Stable	69	226
SK018	Face Oil	Trtmt	1,999	2,000	1,853	1,826	-1.5%	Declining	73	236
SK019	Clay Mask	Mask	1,647	1,867	1,936	2,133	+10.2%	Growing	69	220
SK020	Hydrating Mist	Hydr	1,798	1,792	2,009	2,061	+2.6%	Growing	80	236
TOTAL	**—**	**—**	**36,400**	**36,156**	**37,982**	**38,399**	**+1.1%**	**—**	**—**	**—**

5. Stockout Intelligence & Lost Revenue Analysis

A critical finding from this analysis is the magnitude of revenue lost due to stockouts. Over the 3-year period (2023–2025), **11,762 units of demand went unfulfilled**, representing **$344,192 in lost revenue**. Every SKU experienced stockouts in at least 60% of months — indicating a systemic inventory under-investment problem, not isolated incidents.

SKU	Product	Stockout Rate	Fill Rate	Lost Units (3yr)	Lost Revenue ($)	SafeSt	ROP
SK008	Body Lotion	80.6%	88.6%	678	$18,909	72	230
SK002	Moisturizer	77.8%	86.7%	767	$24,651	69	225
SK010	Eye Cream	77.8%	89.2%	591	$17,352	61	209
SK015	CC Cream	77.8%	88.2%	668	$16,069	76	240
SK020	Hydrating Mist	77.8%	87.4%	694	$18,950	80	236
SK013	Face Scrub	77.8%	87.2%	675	$16,594	77	224
SK004	Night Cream	72.2%	89.4%	583	$17,344	71	228

SK009	Lip Balm	72.2%	87.7%	691	$19,175	69	220
SK005	Serum Vit C	72.2%	87.8%	692	$21,260	75	233
SK014	BB Cream	66.7%	91.6%	436	$12,281	75	226
TOTAL	—	—	—	**11,762**	**$344,192**	—	—

6. Seasonality & Monthly Distribution

The portfolio shows strong and consistent seasonal demand patterns. April through June is the peak demand window, with April at the highest average SI of 1.198. February is the trough month (SI=0.761). Planning should buffer additional stock by late Q1 ahead of the spring peak.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Total Fcst Units	2,957	2,441	2,622	3,841	3,744	3,931	2,975	3,158	3,018	2,708	3,446	3,562
Portfolio SI	0.922	0.761	0.821	1.198	1.164	1.230	0.935	0.990	0.945	0.842	1.079	1.112
Signal	Avg	Low	Low	Peak	Peak	Peak	Avg	Avg	Avg	Low	High	High

7. Strategic Recommendations

Address Systemic Stockout Problem (All SKUs)
Every SKU had stockout rates of 60-80% over 3 years. This is not a forecasting problem — it's an inventory investment problem. Implement the calculated Safety Stock and Reorder Points immediately. Estimated recoverable revenue: $344,192 per cycle (3yr total).

Priority Focus: Moisturizer & Serum Vit C (Lost Rev >$21K each)
SK002 and SK005 represent the highest individual lost revenue at $24,651 and $21,260. Both are high-frequency SKUs.
Increase safety stock to ROP levels and review supplier lead times.

Invest Behind Growth Leaders: Sunscreen, Face Toner, Clay Mask, BB/CC Cream
These 5 SKUs show strongest growth signals (+6% to +10.5%). Prioritize stock availability and ensure marketing support is timed to seasonal peaks (Apr-Jun window).

Prepare for Apr-Jun Demand Surge Across Portfolio
Portfolio SI rises to 1.198-1.230 in April-June. Replenishment orders must be placed by late February / early March to ensure full stock availability at peak season. February is the trough — use it to build buffer inventory.

Monitor Declining SKUs: Lip Balm, Eye Cream, Face Oil
SK009 (-7.2%), SK010 (-10.1%), SK018 (-1.5%) show consecutive demand decline. Investigate root causes (pricing, competition, reformulation needed). Consider reducing safety stock and running promotional campaigns to clear potential excess inventory.

8. Key Assumptions & Limitations

A1 — Portfolio Stability: No new SKU launches or discontinuations in 2026. Portfolio held at 20 SKUs.

A2 — Market Conditions: Competitive landscape and consumer trends broadly stable relative to 2025.

A3 — Seasonality Consistency: 2023–2025 seasonal patterns are representative of 2026. No climate or trend shifts.

A4 — Lead Times: Lead time = 1 month assumed for all SKUs. ROP must be recalculated with actual supplier lead times.

A5 — Price Stability: Unit prices held constant at 3-year average. No promotional pricing or inflation modeled.

L1 — History Depth: Only 3 years of history. More data would substantially improve seasonality and trend estimates.

L2 — Demand Accuracy: Demand Qty may itself be an estimate. Model assumes accurate demand capture during stockouts.

L3 — External Factors: Promotions, competitor launches, viral trends, and ingredient news not modeled.

This forecast was produced using demand-corrected blended modelling (WMA + Holt's DES). It is intended for internal supply chain and commercial planning. All forecasts should be reviewed at the start of Q2 and Q3 as actuals accumulate. For methodology queries contact the Demand Planning Analytics team.